IGCSE 0607 Practice Questions — E4.1 — Combined Questions & Markscheme

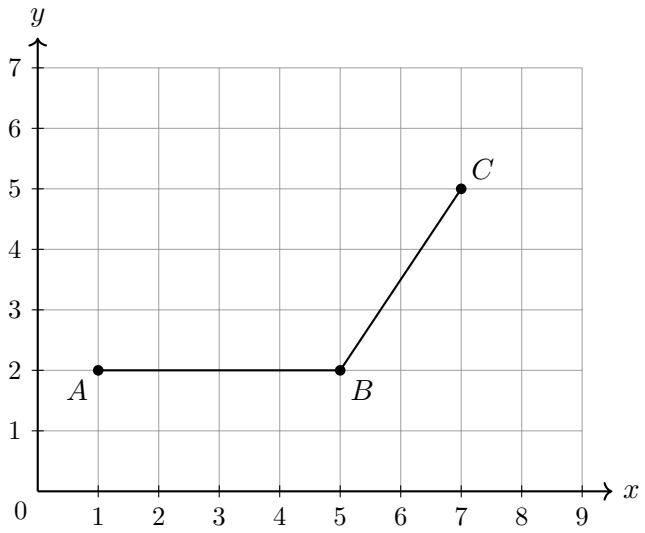
Coordinates

Total Questions: 12 — **Total Marks:** 81 Marks

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- **E4.1** Coordinates
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

1 The diagram shows three vertices, $A(1, 2)$, $B(5, 2)$ and $C(7, 5)$, of a parallelogram $ABCD$.



(a) Find the coordinates of vertex D.

((..... , ...)) **[2]**

(b) Calculate the area of the parallelogram $ABCD$.

..... **[2]**

2 Points $P(-4, 2)$ and $Q(6, 2)$ are plotted on a Cartesian grid.

(a) Write down the coordinates of the midpoint of the line segment PQ.

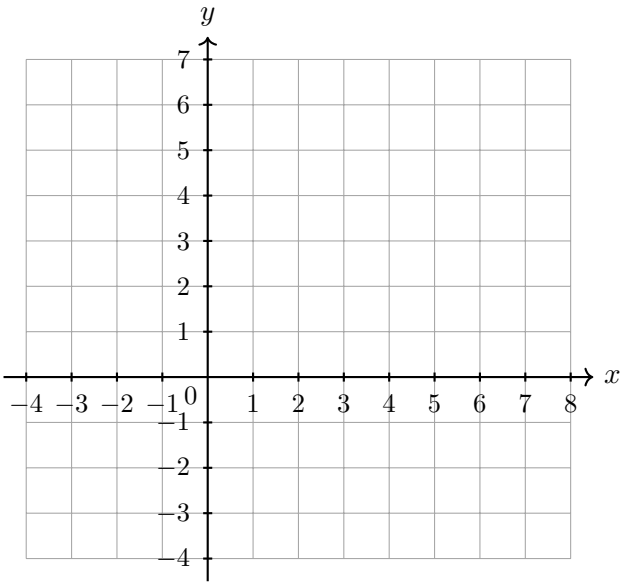
(..... , ...) **[1]**

(b) Point R lies on the positive y-axis.
The area of triangle PQR is 35 square units.
Find the coordinates of R.

(..... , ...) **[3]**

3 The points $A(-1,3)$, $B(2,-1)$ and $C(6,1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

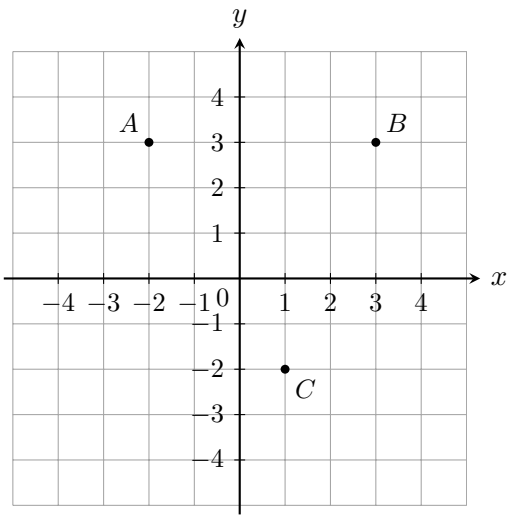
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

4 The diagram shows three points, A, B, and C, plotted on a Cartesian coordinate grid.



(a) Write down the coordinates of point A and point C.

$A =$ ..... , $C =$ ..... **[2]**

(b) Write down the coordinates of point D such that $ABCD$ is a parallelogram.

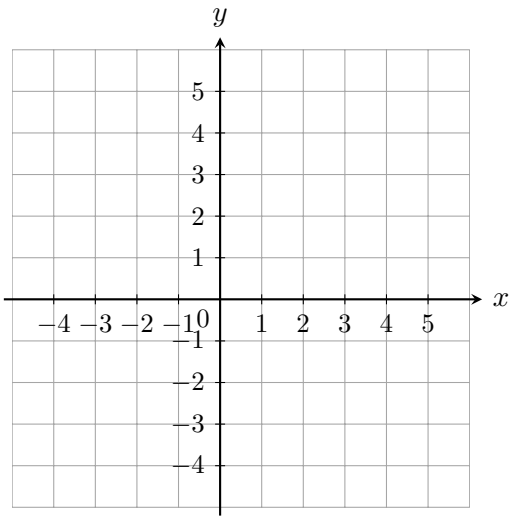
$D =$ (..... , ...) **[2]**

(c) Calculate the area of parallelogram $ABCD$.

..... square units **[2]**

5 In a Cartesian coordinate plane, a kite $PQRS$ has vertices $P(1,5)$, $Q(4,2)$, $R(1,-4)$, and $S(-2,2)$.

(a) Plot the points $P(1,5)$, $Q(4,2)$, $R(1,-4)$, and $S(-2,2)$ on the grid below and join them in order to form the kite $PQRS$.



[2]

(b) Write down the coordinates of the point where the diagonals of kite $PQRS$ intersect.

(..... , ...) **[1]**

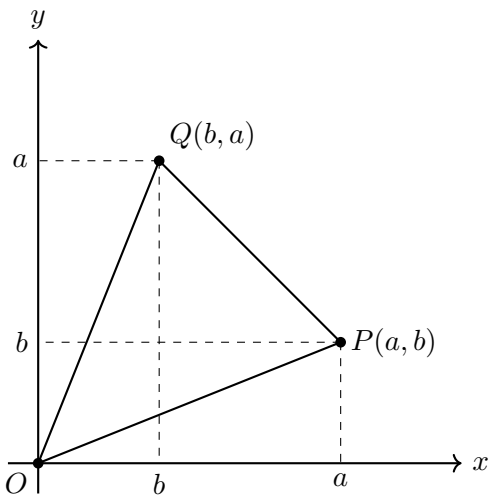
(c) Calculate the area of kite $PQRS$.

..... square units **[2]**

(d) Point P is reflected in the x-axis to give point P', and point Q is reflected in the y-axis to give point Q'.
Write down the coordinates of P' and Q'.

$P' =$ ..... , $Q' =$ ..... **[2]**

6 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

7 A rectangle has its sides parallel to the coordinate axes.
Two opposite vertices of the rectangle are $A(-2, -4)$ and $C(x, y)$, where $x > 0$ and $y > 0$.
The area of the rectangle is 54 square units and its perimeter is 30 units.

(a) Show that $x + y = 9$.

[2]

(b) Form an equation in x and solve it to find the two possible pairs of coordinates for C.

[4]

(c) For the case where $x > y$, write down the coordinates of the other two vertices of the rectangle.

[2]

8 Points A and B have coordinates $(3k, -k)$ and $(k+4, -5k)$ respectively, where k is a constant.

(a) The midpoint of the line segment AB lies on the y-axis.
Find the value of k.

$k =$ ..... **[2]**

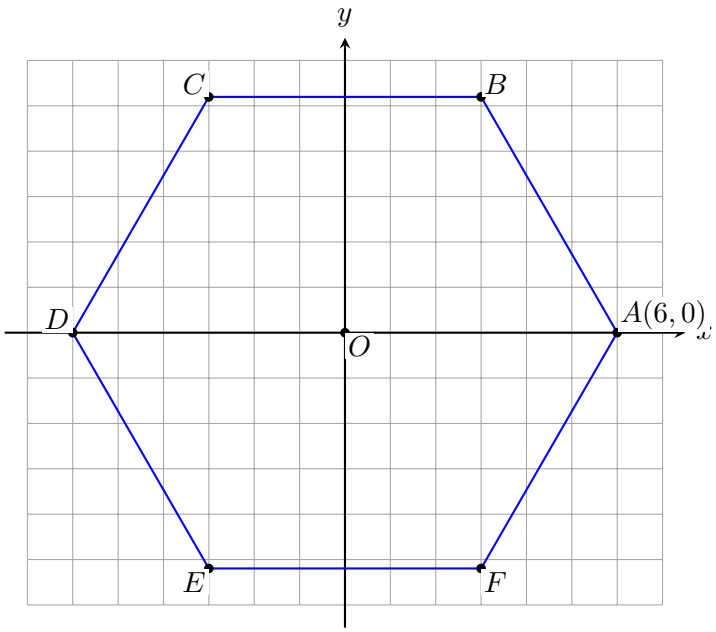
(b) Using your value of k from part (a), find the exact length of AB.
Give your answer in the form $a\sqrt{b}$ where b is a prime number.

..... **[3]**

(c) A point C lies on the x-axis.
The distance AC is equal to the distance BC.
Find the coordinates of C.

(..... , ...) **[4]**

9 A regular hexagon $ABCDEF$ has its center at the origin $O(0, 0)$.
The point A has coordinates $(6, 0)$.



(a) Write down the coordinates of point D.

(..... , ...) **[1]**

(b) Calculate the coordinates of point B.
Give your values correct to 2 decimal places.

(..... , ...) **[3]**

(c) Point P lies on the line segment AD such that $AP : PD = 1 : 3$.
Find the coordinates of P.

(..... , ...) **[2]**

10 A straight line has the equation $2y + 3x = 12$.
The line intersects the y-axis at A and the x-axis at B.
The point O is the origin $(0, 0)$.

(a) Find the coordinates of A and B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Point C is such that $OACB$ is a rectangle.
Write down the coordinates of C.

((..... , ...)) **[1]**

(c) Point P lies on the line segment AB and has an x-coordinate of 2.
Find the coordinates of P.

((..... , ...)) **[2]**

(d) Find the area of triangle OPB.

..... **[2]**

11 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]

12 A rectangle $ABCD$ has vertices $A(-3,-1)$, $B(5,-1)$, $C(5,k)$, and $D(h,k)$, where $k>0$.

(a) State the value of h and the quadrant in which point D lies.

$h =$ ..... , Quadrant: ..... **[2]**

(b) Given that the area of rectangle $ABCD$ is 52 square units, calculate the value of k.

$k =$ ..... **[3]**

(c) Point M is the centre of rectangle $ABCD$.
Find the coordinates of M.

$M =$ (..... , ...) **[2]**

(d) Point $E(p,q)$ lies on the y-axis such that triangle ABE has an area of 18 square units and $q>0$.
Find the coordinates of E.

$E =$ (..... , ...) **[3]**

1 **(a)** x-coordinate 3 **[B1]**

y-coordinate 5 **[B1]**

(b) Base $AB = 4$ and height $= 3$ **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $(1, 2)$ **[B1]**

(b) Identifies the base length $PQ = 10$ **[M1]**

Sets up the area equation:

$$\frac{1}{2} \times 10 \times h = 35$$

h $= 7[M1]$

$(0, 9)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $A = (-2, 3)$ **[B1]**

$C = (1, -2)$ **[B1]**

(b) Horizontal shift of 5 units or vertical shift of 5 units identified **[M1]**

$(-4, -2)$ **[A1]**

(c) Base $= 5$ and perpendicular height $= 5$ **[M1]**

25 **[A1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** All four points P, Q, R, and S plotted accurately **[B1]**

Points joined with straight line segments to form kite $PQRS$ **[B1]**

(b) $(1, 2)$ **[B1]**

(c) Diagonal lengths $PR = 9$ and $QS = 6$ **[M1]**

$\frac{1}{2} \times 9 \times 6 = 27$ **[A1]**

(d) $P' = (1, -5)$ **[B1]**

$Q' = (-4, 2)$ **[B1]**

[Back to Question] *[Total Marks: 7]*

6 (a) Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

$2(a-b)^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

$a^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Back to Question] *[Total Marks: 8]*

7 (a) Width $= x - (-2) = x + 2$ and Height $= y - (-4) = y + 4$ **[M1]**

$2(x+2) + 2(y+4) = 30 \implies 2x + 2y + 12 = 30 \implies x + y = 9$ **[A1]**

(b) Area $= (x+2)(y+4) = 54$ **[M1]**

Substitute $y = 9 - x$ to give $(x+2)(13-x) = 54$ **[M1]**

$x^2 - 11x + 28 = 0$ or equivalent three-term quadratic **[M1]**

$(4, 5)$ and $(7, 2)$ **[A1]**

(c) $(-2, 2)$ **[B1]**

$(7, -4)$ **[B1]**

[Back to Question] *[Total Marks: 8]*

8 (a) Sets the x-coordinate of the midpoint to zero: $\frac{3k + k + 4}{2} = 0$ **[M1]**

-1 **[A1]**

(b) Substitutes $k = -1$ to find $A(-3, 1)$ and $B(3, 5)$ **[M1]**

$AB^2 = (3-(-3))^2 + (5-1)^2 = 6^2 + 4^2 = 52$ **[M1]**

$2\sqrt{13}$ **[A1]**

(c) Lets C be $(x, 0)$ and sets up equation $AC^2 = BC^2$:
$(x+3)^2 + (0-1)^2 = (x-3)^2 + (0-5)^2$ **[M1]**

Expands both sides correctly: $x^2 + 6x + 9 + 1 = x^2 - 6x + 9 + 25$ **[M1]**

Rearranges to solve for x: $12x = 24$

x = 2[$M1$]

$(2, 0)$ **[A1]**

[Back to Question] *[Total Marks: 9]*

9 (a) $(-6, 0)$ **[B1]**

(b) Identifies radius $OB = 6$ and angle from positive x-axis as $60°$ **[M1]**

$x = 6\cos 60° = 3$ and $y = 6\sin 60°$ $(= 3\sqrt{3})$ **[M1]**

$(3, 5.20)$ **[A1]**

(c) Calculates length $AD = 12$, and finds $AP = \frac{1}{4} \times 12 = 3$ **[M1]**

$(3, 0)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 (a) $A(0, 6)$ **[B1]**

$B(4, 0)$ **[B1]**

(b) $(4, 6)$ **[B1]**

(c) $2y + 3(2) = 12$ **[M1]**

$(2, 3)$ **[A1]**

(d) Base $OB = 4$ and height of $\triangle OPB = 3$ (the y-coordinate of P) **[M1]**

6 **[A1]**

[Back to Question] *[Total Marks: 7]*

11 (a) Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. x-coordinate $= 6 + \frac{2}{3}(-6)$ OR
y-coordinate $= 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** $h = -3$ **[B1]**

Second quadrant (accept 2nd or II) **[B1]**

(b) Width $AB = 5 - (-3) = 8$ **[M1]**

$8(k + 1) = 52$ or $k + 1 = 6.5$ **[M1]**

5.5 **[A1]**

(c) $x = \dfrac{-3 + 5}{2} = 1$ or $y = \dfrac{-1 + 5.5}{2} = 2.25$ **[M1]**

$(1, 2.25)$ **[A1]**

(d) $p = 0$ **[B1]**

$\dfrac{1}{2} \times 8 \times (q + 1) = 18$ or $q + 1 = 4.5$ **[M1]**

$(0, 3.5)$ **[A1]**

[Back to Question] *[Total Marks: 10]*